Solutions and teaching notes for chapter 8

Teacher use. The answers below concern the tasks not worked in the lesson. First have students state their method; give graded help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. 4. *Action:* Add, then divide by the number of values.
2. Yes: positive numbers with sum 1. *Action:* Put over a common denominator.
3. 9. *Action:* Keep the brackets before squaring.
4. 0.4. *Action:* Use the complement.
5. 5. *Action:* Write net gain = received minus paid.
6. 0.5. *Action:* Check the square of the answer.

Investigation activity

Gains $(-1, 0, 1)$ and $(-3, 0, 3)$: expectation 0 in both cases, variances $\frac{2}{3}$ and 6. In standard deviation, the second is three times more dispersed.

Immediate applications

1. Values 0, 1, 2; probabilities $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{4}$. $P(X \leq 1) = \frac{3}{4}$.
2. $E(G) = \frac{10}{5} = 2$. For a stake m, $E(G - m) = 2 - m$; fair stake 2.
3. $E(X) = 0$, $E(X^2) = 1$, variance 1 and standard deviation 1.
4. $E(Y) = -1$, $V(Y) = 16$, $\sigma(Y) = 4$.
5. $m = \frac{6}{5} = 1.2$ and $E(X) = 1$: the gap is 0.2 for this sample.

Exercises: method and results

Exercise 8.1 – A distribution to build. Values 0, 1, 2; each obtained from two faces, probabilities $\frac{1}{3}$. $P(X < 2) = \frac{2}{3}$. Beware the remainder 0 for 3 and 6.

Exercise 8.2 – Two variance calculations. $E(X) = -\frac{1}{2} + 1 = \frac{1}{2}$. $E(X^2) = 1 + 4 = 5$; $V = 5 - \frac{1}{4} = \frac{19}{4}$, $\sigma = \frac{\sqrt{19}}{2}$. Direct: $(\frac{25}{4})(\frac{1}{4}) + (\frac{1}{4})(\frac{1}{2}) + (\frac{49}{4})(\frac{1}{4}) = \frac{76}{16} = \frac{19}{4}$.

Exercise 8.3 – A fair game. $E(G) = \frac{15}{5} = 3$, so m=3. $E(G^2) = \frac{125}{5} = 25$; $V(G) = 25 - 9 = 16$, standard deviation 4, unchanged by the stake.

Exercise 8.4 – Same mean, different risk. $E(X) = E(Y) = 1$; $V(X) = 0$; $E(Y^2) = 10$, $V(Y) = 9$. X is certain, Y is dispersed. A player's preferences cannot be ranked from the expectation alone.

Exercise 8.5 – Linearity without independence. $E(X) = \dfrac{1+2+3+4+5+6}{6} = \dfrac{7}{2}$; $E(Y) = 7 - E(X) = \dfrac{7}{2}$. Since $X + Y = 7$, its expectation is 7, the sum of the two expectations. The events $\{X = 1\}$ and $\{Y = 6\}$ are identical: their intersection has probability $\dfrac{1}{6}$, whereas the product of their probabilities is $\dfrac{1}{36}$. The variables are not independent. Linearity remains valid without this hypothesis.

Exercise 8.6 – A minimal loss. Write $X - a = (X - \mu) + (\mu - a)$, expand, then use $E(X - \mu) = 0$. The square $(a - \mu)^2$ is minimal, zero only for $a = \mu$. The mean minimises the mean squared error.

Exercise 8.7 – Letter frequencies. Convert the text to lower case, keep the characters for which `isalpha()` is true, then count the "a"s. Divide this number by the number of letters if it is nonzero. For a text with no letter, return `None`, because the frequency is not defined. For `Aba !`, two "a"s out of three letters give $\dfrac{2}{3}$; the other two tests give `None`. The supplied program `frequence_a` follows this convention.

Exercise 8.8 – Delayed review. Distribution: $0, 1, 2$ with probabilities $\dfrac{1}{4}, \dfrac{1}{2}, \dfrac{1}{4}$. Expectation 1; second moment: $\dfrac{1}{2} + 1 = \dfrac{3}{2}$, variance $\dfrac{1}{2}$. The four paths are counted explicitly.

CQFD, exercise 28 p. 306, all questions. $E(X) = (-10) \times 0.35 + (-5) \times 0.15 + 0 \times 0.4 + 20 \times 0.1 = -2.25 = -\dfrac{9}{4}$. A negative expectation is compatible with a possible positive value.

CQFD, exercise 31 p. 306, all questions; use the distribution of exercise 28. $E(X^2) = 100 \times 0.35 + 25 \times 0.15 + 400 \times 0.1 = 78.75$. Thus $V(X) = 78.75 - (-2.25)^2 = 73.6875 = \dfrac{1179}{16}$, then $\sigma(X) = \dfrac{\sqrt{1179}}{4} \approx 8.6$. Do not round the expectation before calculating the variance.

MCQ: answers and reasons

MCQ 1.

1. B. Partition of the sample space.
2. A. $0 + \dfrac{4}{4}$.
3. D. König–Huygens.
4. C. Positive root.
5. B. Linearity.

MCQ 2.

1. C. Translation with no new dispersion.

2. A. Absolute value of the factor.
3. D. A variable can be negative or have a mean that is never attained.
4. B. A common finite sample space is enough.
5. C. $V(X) = E(X^2) - E(X)^2 = 5 - 4 = 1$.

True or false

1. False: it is a mean of squares with positive weights.
2. False: the constant 1 and an equally likely variable taking $-2, 4$ have the same mean.
3. True: the deviations from the new mean stay the same.

Differentiated teaching. For a technical block, offer only the first step (domain, factor or reference sample space), then withdraw the help. For an independent student, ask for a second method and the treatment of a limiting case. Orally, allow an initial reformulation in the language the student has mastered, then require a precise mathematical sentence in French.